

Early-Stage Clinical Opportunity Assessment

Lung Entrapment by Organized Pleural Rind in Benign Pleural Disease

A Needs-Driven Assessment of Device and Biologic Intervention Routes

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BME 501 — MedTech Innovation & Entrepreneurship
Deliverable 9 — Final Report
July 2026

1. Executive Summary

BioDesign Need Statement

A way to prevent or reverse entrapment of the lung by an organized fibrous rind, in patients with benign pleural disease, in order to restore and preserve re-expansion of the lung against the chest wall. A patient who undergoes chest-tube drainage and intrapleural enzymatic therapy may progress to an organized collagenous rind that mechanically entraps the lung, producing permanent restrictive dysfunction and referral for decortication — an operation carrying 3.1% mortality and a 39.3% complication rate [3] — or, if the patient is not a surgical candidate, no definitive option at all. A solution is needed that restores re-expansion without thoracotomy where fibrinolytics have failed, achieving radiographic re-expansion in $\geq 60\%$ of treated patients with a safety profile permitting use in patients declined for surgery.

Three sources converge. A multicenter analysis of 466 patients found real-world tPA/DNase resolution of 78%, with pleural thickening ranked first or second among failure predictors in all four models tested [30]; the thoracic surgeon interviewed identified the fibrous rind unprompted as the reason lytic therapy fails and the reason the operation is mechanically brutal [Interview 1]; and the sponsor of the leading competitor drug reported that patients requiring surgery for evolving fibrosis or trapped lung "were not well captured by the protocol definition" of treatment failure [10]. That trial excluded fibrothorax by protocol and concluded that efficacy in organizing effusions remains unestablished [10] — the organizing-rind population is not served by the drug closest to market. Alternatives run out at exactly this point: antibiotics and drainage for Stage I, enzymatic therapy for Stage II [29, 38], VATS or open decortication for Stage III [2, 8], and nothing for non-surgical candidates. The unmet outcomes are re-expansion in the $\sim 22\%$ who fail enzymatic therapy [30], elimination of the tearing force of rind stripping and its 7.5% prolonged air leak [5], and any definitive option for the subgroup that cannot be operated on, where half of patients allocated to surgery in a bias-corrected trial never received it [37]. Adoption requires delivery through an existing chest tube or standard ports, no capital equipment, bleeding not exceeding the 4.2% observed across 1,850 tPA/DNase patients [36], and ≤ 30 minutes added procedure time.

The barriers are real: decortication works well in experienced hands, both routes are bundled into a fixed DRG, the biologic faces a 7–10 year BLA with no pleural collagenase predicate, and the device faces predicate exposure to cleared hydrodissection technology. The US decortication denominator is derived rather than counted — the parapneumonic funnel supports 2,000–12,000 procedures annually and the $\sim 10,000$ used here sits at the top of that range [2, 7]. At \$550 ASP the decortication market is \$5–6M, rising to \$8.1–8.8M with severe adhesiolysis [20, 22]. For the biologic, 40,000 pleural drainage patients annually [11] at 50% enzymatic failure gives 20,000 rescue patients; preventing surgery in 40% avoids 8,000 decortications at $\sim \$20,000$ each [33], a \$96M serviceable market at 60% capture and a \$4,800 ASP. Water-jet hydrodissection is a cleared mechanism with a plausible 510(k) in 6–12 months, IP exposure being the dominant risk; collagenase is mechanistically correct but has no pleural predicate, and penetration and selectivity are unproven. The recommendation is to proceed to prototyping and clinician co-creation on the device route while holding the biologic as a parallel research track. Investment is two-tiered: \$2,000–\$8,900 executable within the program — the pulmonology interview, a patent landscape search, ex vivo rind separation and collagenase penetration tests, and an HCUP/NIS pull to convert the derived denominator into a counted one — gating \$635K–\$1.16M in Tier 2 for formal FTO opinion, animal model, prototype and pilot study. Proceed if FTO is clear, ex vivo testing shows lower visceral pleural injury than blunt dissection, counted volume supports $\geq 8,000$ decortications annually, and a reimbursement path is identified.

2. Introduction

2.1 Background and Strategic Context

Empyema is the most common complication of pneumonia, and its incidence in the United States has been rising since the 1990s [2]. Mummadi et al., analyzing national data, documented a 37.5% increase in pleural infection hospitalizations between 2007 and 2016, with a mean length of stay of 13.8 days and mean cost per admission of \$38,591 [1]. Despite antibiotics, image-guided drainage, and intrapleural enzymatic therapy, a substantial fraction of these patients still arrive at an operation.

The disease is conventionally staged in three phases. Stage I is inflammatory, with free-flowing fluid that drains with a tube. Stage II is fibrinopurulent, with fibrin septation and loculation — this is where enzymatic therapy acts. Stage III is organizing: fibroblasts populate the fibrinous matrix and deposit cross-linked collagen, forming a rigid peel that encases the lung [8]. Transforming growth factor-beta drives collagen synthesis by mesothelial cells in this transition [42].

The clinical significance of this staging is that the therapeutic mechanism changes underneath it. Fibrinolytics cleave fibrin. Once the matrix has been remodeled into collagen, the drug and the target no longer match. This is not a dosing problem or a delivery-volume problem; it is a substrate problem. The practical consequence is that a patient who crosses from Stage II to Stage III during a course of enzymatic therapy has, in effect, aged out of the treatment while receiving it.

Two features of this process matter for solution design. The first is convergence: the same fibrotic endpoint is reached from several distinct benign triggers. Organized haemorrhagic effusion, tuberculous effusion, empyema, and asbestos-related pleurisy all terminate in excessive matrix deposition that destroys normal pleural architecture and compromises function [41]. This is what justifies scoping the need across benign pleural disease rather than empyema alone.

The second is how well the mechanism is actually understood. While a 2004 review still considered the underlying causes and key cellular players to be debated [41], the field has since mapped the pathway with precision [46]. We now understand that the transition involves 'mesomesenchymal transition,' where pleural cells transform into scar-producing myofibroblasts. This process is driven by specific chemical signals (such as TGF-beta) that instruct the cells to begin rapidly depositing collagen and losing their structural integrity. In short, the biological transition from a manageable fluid collection to a rigid, collagen-based scar is no longer a 'black box'—the pathway is now clearly understood.

However, a gap remains. While we understand how the scar forms, we do not fully grasp the underlying mechanics—specifically, where these scar-producing cells originate or why they persist once fibrosis is established [46]. This is critical: any therapy aimed at reversing this damage must move beyond merely describing the formation pathway and instead identify what determines whether the scar resolves or endures.

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Decortication reverses this mechanically, and it works: the operation restores measurable pulmonary function, with one series reporting FVC improving from 69.74% to 85.40% predicted post-decortication [6]. But it is a

major thoracic operation with real cost. In the STS General Thoracic Surgery Database analysis of 7,316 decortications, operative mortality was 3.1%, complications occurred in 39.3% of patients, and major morbidity in 15.6% [3]. And a meaningful subset of patients — the elderly, the frail, the physiologically marginal — are never offered it.

2.2 Project Objectives

This assessment sets out to answer four questions:

- Is organized pleural rind a real and distinct clinical need, separable from the drainage/loculation need that existing therapies already address?
- How large is the affected population, and how confident can we be in that number?
- What would a solution have to do, and what would it have to avoid doing, to be adopted by the clinicians who currently manage these patients?
- Which of two candidate solution routes — a device that makes rind removal safer, or a biologic that dissolves rind without an operation — represents the better risk-adjusted opportunity?

A deliberate scope decision was made early and held throughout: malignant trapped lung is excluded from this need. Malignant pleural disease is managed palliatively with indwelling catheters, and including it would both dilute the population and undermine the central claim that surgery is currently the only definitive option.

2.3 Methodology Overview

Primary Research

Interview 1 — Thoracic Surgeon (Dr. Cohen). Semi-structured interview covering the mechanics of decortication, the point at which patients are referred to surgery, the range of etiologies producing an entrapped lung, and the intraoperative experience of removing rind. Anchored to specific recent cases rather than general impressions.

Interview 2 — Cardiothoracic ICU Physician Assistant. Semi-structured interview covering the intrapleural enzymatic therapy protocol as actually executed, who performs each step, dose adherence, failure patterns, and workflow hazards. This interview was designed to test — and ultimately falsified — an earlier hypothesis that nursing labor burden was the core need.

Observation. Direct observation of one decortication procedure, providing firsthand exposure to the mechanical character of rind removal and the intraoperative decision points.

Pending. A pulmonology / interventional pulmonology interview remains outstanding and is the highest-priority gap. While the CTICU perspective offers a biased sample restricted to the failure tail, pulmonology encompasses the true patient denominator and retains the definitive clinical judgment on whether an enzymatic course is effective.

Secondary Research

Literature was drawn from MEDLINE and journal sources across five domains: epidemiology and disease burden [1, 2, 7, 11, 44]; staging distribution and natural history [8, 25, 26, 27]; enzymatic therapy outcomes and failure prediction [28, 29, 30, 31, 36, 37, 38, 39]; surgical outcomes, morbidity, and conversion [3, 5, 6, 12, 13, 23, 24, 28]; and health economics [1, 14, 15, 16, 17, 32, 33, 34]. Regulatory and coding research drew on CPT descriptors [43]. Competitive intelligence centered on the LTI-01 program [9, 10].

Analytical Framework

The project utilized the Stanford Biodesign methodology as its structural framework. This process involved generating a solution-agnostic need statement, scoping the clinical need, and performing systematic filtering via a 5/3/1 rubric. The rubric scored potential solutions based on their mechanism of action, the size of the affected patient population, the severity of clinical consequences, and the ease of demonstrating clinical proof. To estimate the overall value of the need, the problem cost per case was multiplied by the total population and the projected outcome improvement. Market forecasting was conducted using bottom-up procedure volumes integrated with a staged adoption curve, which was informed by the adopter segmentation from the diffusion-of-innovation model [18].

3. Clinical Landscape

3.1 The Parapneumonic Funnel

The population is best understood as a funnel with substantial attrition at each stage, and with genuine uncertainty in the width of each band. One figure in it is firmer than the rest: the PILOT study, the largest international prospective study observational cohort in pleural infection, found that standard medical therapy with a chest tube and antibiotic fails in 33.5% of cases [45].

Stage of funnel	Estimate	Source and confidence
US pneumonia hospitalizations / yr	~1,000,000	AATS consensus [2]. Widely repeated; order-of-magnitude reliable.
Develop parapneumonic effusion	20–40%	AATS consensus [2]. Broad range reflects heterogeneous definitions.
Progress to empyema	5–10% of effusions	AATS consensus [2].
Serious pleural infections / yr (US)	~90,000	Cited in LTI-01 Phase 1 [9]. Secondhand citation; not independently verified.
Require pleural drainage / yr (US)	~40,000	Tucker et al. [11].
Require surgical drainage	~20% (IQR 1–32%) / ~30%	Systematic review of 227,898 patients [7]; AATS states ~30% [2]. The two disagree.
Decortications captured, STS 2009–2016	7,316 across 263 sites	STS GTSD [3]. A participating-institution subset, not a national count.

3.2 Stage Distribution at Presentation

Three independent surgical series give a sense of how patients distribute across stages at the time they reach an operation — though each is a surgical cohort and therefore already enriched for advanced disease:

- A cohort of 81 primary empyema patients: 36% Stage II, 64% Stage III [25].
- A series of 274 patients: Stage I in 78, Stage II in 157, Stage III in 39 [27].
- A cohort of 719 surgically treated Stage II–III patients: 76.9% Stage II, 23.1% Stage III [26].

The variability across these series is itself informative — staging is applied inconsistently, and the boundary between 'late Stage II' and 'early Stage III' is a radiographic and intraoperative judgment rather than a measurement. This matters directly for any product whose indication would be stage-defined: the inclusion criterion for a trial in this space cannot simply say 'Stage III' without an objective operational definition.

What is consistent is the consequence of stage on surgical difficulty. In the 719-patient series, conversion from thoracoscopic to open approach occurred in 8.4% of Stage II cases and 19.2% of Stage III cases [26] — a more than twofold increase attributable to the maturity of the peel.

3.3 Why Enzymatic Therapy Fails

The optimistic figure circulating in reviews is that roughly 90% of patients are cured with tPA/DNase — but the source reporting that figure does not stratify by stage, which is precisely the variable that determines whether the drug can work [28]. The more defensible number comes from a multicenter machine-learning analysis of 466 patients across five US institutions, where real-world resolution was 78%, and where pleural thickening and abscess/necrotizing pneumonia were the dominant predictors of failure across all four models tested [30]. A recent review similarly places enzymatic treatment failure at up to 25% [38].

Pleural thickening is the radiographic signature of the organizing rind. That it emerges as a top predictor — while loculation, the thing the drug is designed to dissolve, does not — is the single most important finding supporting this need statement. The therapy fails where the substrate has changed.

There is currently no way to know this in advance. As one review of the field states plainly, existing biochemical tests and imaging cannot assess a patient's fibrinolytic system or predict whether they will respond to enzymatic therapy versus require surgery [31].

Nor is the protocol itself settled. In a position statement, the two figures most central to this field — the investigator behind the intrapleural fibrinolytic research program and the investigator who led the MIST trials — write that roughly seventy years of fibrinolytic therapy, which agent to use, and how to combine a fibrinolysin with DNase, and the optimal dose and dosing interval all remain unresolved. They call specifically for fundamental research to enable personalized therapy or bedside dosing guidance [39]. The six-dose, twelve-hourly, three-day regimen delivered at the interview site is therefore not an evidence-derived optimum; it is a convention.

The CTICU interview added a clinical observation that the literature has not explained: response is bimodal. Enzymatic therapy either 'worked really well or didn't work at all,' and which of the two is often apparent after the first dose [Interview 2]. If that observation generalizes, the standard six-dose, twelve-hourly, three-day protocol is committing most patients to two additional days of a treatment whose outcome was already determined — days that matter, because surgical morbidity and mortality rise once preoperative delay exceeds five days [3].

Whether those days carry a real cost is genuinely unsettled. On one side, the STS database shows surgical morbidity and mortality rising once preoperative delay exceeds five days [3], which would make time lost to a futile enzymatic course time that worsens the operation the patient ultimately needs. On the other, MIST-3 randomized patients to early enzymatic therapy or early surgery and achieved a significant separation in time to intervention — a median of 1.0 versus 3.5 days ($P = 0.02$) — without producing any significant difference in length of stay: 7 days in each intervention arm against 10 days for standard of care, 9 days overall, $P=0.70$ across all three groups [37]. Not only did the two intervention arms matched each other; but the apparent three-day advantage over standard care did not reach significance either. MIST-3 was a feasibility trial and was not powered for length of stay, so this is an absence of evidence rather than the evidence of absence — but it is the only randomized evidence available, and it does not support a timing claim in either direction.

The two findings are not strictly contradictory. A 2.5-day separation inside the first week of treatment is not the same event as crossing a five-day preoperative threshold, and the STS cohort is far larger. But together, they bound how much weight a speed-based argument can carry. A solution justified primarily on saving stands on weaker ground than one justified on removing a substrate that surgery is currently the only way to address — which is an argument for positioning any intervention developed here against the rind itself rather than against the clock.

3.4 Consequences of Failure to Resolve

The Population That Cannot Have the Operation

The size of the non-surgical-candidate population is normally invisible, because surgical series are drawn from patients already selected as operable: patients who undergo surgery in observational cohorts are consistently younger and carry fewer comorbidities than unselected pleural infection populations [37]. MIST-3 is the exception. It randomized patients to a surgical arm regardless of fitness for surgery, specifically to eliminate the selection bias of earlier surgical trials. Only 10 of 20 patients allocated to VATS actually received it. Anaesthetic risk alone accounted for 2 of 20 and surgical capacity for another [37].

In a trial deliberately designed to evaluate the true surgical potential across an unselected population, half the allocated patients did not get the operation. That is the clearest published quantification of the gap this statement targets.

Primary research suggests that gap may also be wider than the trial data implies, and different in composition. The interview site serves a patient population with substantial injection drug use [Interview 1]. Patients who inject drugs are frequently poor surgical candidates for reasons unrelated to age or frailty: ongoing substance use, reinfection risk, and concerns about postoperative adherence and follow-up. If the non-surgical-candidate population includes younger patients declined on those grounds as well as the elderly and comorbid patients that MIST-3 and the STS data describe, then the target population is larger and demographically different from the frailty-driven picture drawn above.

Mortality

One-year mortality in pleural infection is approximately 20%, rising to as high as 35% in frail and elderly patients [4]. It is important to attribute this correctly: this mortality belongs to active organizing empyema — an ongoing infection with systemic consequences — instead of chronic trapped lung, which represents a stable mechanical condition that degrades a patient's quality of life and necessitates ongoing hospital visits for fluid drainage and persistent breathlessness.

The mortality differential between operative and nonoperative management is large and consistent across two independent datasets. In a cohort of 9,014 hospitalized patients, nonoperative management was associated with higher mortality at every timepoint: 17.2% versus 10.6% in-hospital, 11.1% versus 4.2% at 30 days, 26.6% versus 15.0% at 6 months, and 32.3% versus 18.8% at one year [23]. A separate analysis found 30-day mortality of 16.6% with nonoperative therapy versus 5.4% with operative therapy, along with higher readmission (5.9% vs 3.0%) and higher discharge to intermediate care facilities (30.1% vs 21.8%) [24].

These comparisons describe two different patient populations rather than the efficacy of the surgery itself. Because nonoperative management is heavily enriched for older, immunocompromised, and comorbid patients who were declined surgery, the mortality gap primarily reflects selection bias.

The MIST-3 trial isolates this bias by randomizing patients to a surgical arm regardless of fitness, analyzed by intention to treat. Its 12-month mortality data ran counter to the observational findings: 20% in the VATS arm (4 of 20) against 5.3% for enzymatic therapy (1 of 19) and 4.8% for standard of care (1 of 21), $P = 0.19$. This evidence demonstrates that the "mortality advantage" observed in prior studies was not a benefit of the surgery, but a result of selecting healthier, stronger patients for operation.

What survives this scrutiny is the finding that matters here. Whether the non-operated population dies at higher rates because they did not receive an operation, or because they were never well enough to receive one, they die at higher rates — roughly a third within a year [23]. That is the population with no definitive option, and it is the population this need statement targets. For present purposes the casual question of how much surgery

would have helped them is less important than the descriptive fact that surgery was not available to them. The selection bias is not an obstacle to the argument; the confounder is the population.

Morbidity of the Alternative

Decortication carries its own burden. Across 7,316 STS-captured procedures: 3.1% mortality, 39.3% complication rate, 15.6% major morbidity, median 4 days preoperative and 7 days postoperative length of stay, 26.3% requiring transitional care after discharge, and 8.7% readmitted within 30 days [3]. Prolonged air leak specifically occurs in 7.5% of empyema decortications [5].

Chronic pain is the under-discussed cost. Chronic intercostal neuralgia occurs in up to 44% of patients at six months post-thoracotomy, and roughly one third of VATS patients still report persistent pain or discomfort at the operative site at three to eighteen months [28]. Conversion from VATS to open is common enough to matter: close to one fifth of VATS procedures convert [28], with one series reporting 44.4% conversion among 18 VATS attempts [13], and a Stage III–focused series reporting only 4.5% [12]. That spread — 4.5% to 44% — reflects case selection and surgeon experience more than technique.

Untreated, the natural history is worse still: pulmonary fibrosis with lung trapping and dyspnea is the most common outcome, alongside chest wall contraction and deformity, empyema necessitans, bronchopleural fistula, osteomyelitis, pericarditis, mediastinal or subphrenic abscess, sepsis, and death [44]. Roughly 25% of patients in the MIST2 population were hospitalized for more than a month [29].

3.5 Critical Needs Criteria and Acceptance Criteria

Criterion	Must Have (Table Stakes)	Nice to Have (HILS Upside)
Efficacy	Radiographic re-expansion in $\geq 60\%$ of enzymatic-failure patients	Re-expansion $\geq 80\%$; measurable FVC recovery toward the $\sim 85\%$ predicted seen post-decortication [6]
Safety	Bleeding rate not exceeding the $\sim 4\%$ seen with tPA/DNase [36]; no bronchopleural fistula	Safe enough for patients formally declined for surgery
Delivery	Through existing chest tube or standard VATS ports	Single administration rather than a multi-day course
Time to effect	Observable change on imaging within 7 days	Within 48 hours, enabling same-admission decision-making
Workflow	≤ 30 min added provider time; no new staffing	Reduces total enzymatic dosing burden
Economics	Priced below the \$15,000 enzymatic course it rescues [32]	Demonstrable LOS reduction against the 13.8-day mean [1]
Compatibility	Concurrent antibiotics; standard drainage hardware	Usable at community hospitals without IP service

4. Primary Research Findings

Two clinician interviews and one procedure observation form the primary data for this assessment. They are reported here in detail because they went beyond merely supplementing the literature; they were foundational to this assessment, fundamentally reshaping the project's direction.

Interview 1 — Thoracic Surgeon

Practicing thoracic surgeon at WakeMed Raleigh; performs decortication. Interviewed as the downstream recipient of enzymatic-therapy failures and as the operator who physically removes rind. A surgeon sees only the failure tail and is therefore a valid source on operative mechanics, referral triggers, and etiology breadth, but not a valid source for incidence or denominator.

Finding 1.1 — The rind is why the drug fails

The surgeon independently identified organized fibrous rind as the reason intrapleural fibrinolytic therapy fails, without being prompted toward that answer. This corroborates the machine-learning finding that pleural thickening dominates failure prediction [30] and was the pivot point for locking the need statement onto tissue rather than drainage.

Finding 1.2 — The operation is mechanically brutal

Rind removal is performed by hard mechanical force. The peel is gripped and stripped off the visceral pleura with instruments including forceps — not delicate dissection, but the controlled application of tearing force to separate a tough collagenous layer from the fragile lung surface beneath it. The surgeon's own verdict on the procedure was unambiguous: "If there were a way to get rid of the rind without surgery, nobody would miss doing decortications". There is no way to strip the peel without pulling on the lung. The two are adherent; the force that frees one acts on the other, and 7.5% of empyema decortications end in prolonged air leak [5]. Whether that force can be made selective is the question the device route exists to answer; whether it can be dispensed with altogether is the question behind the biologic route.

Finding 1.3 — Etiologies beyond infection

The surgeon identified four further routes to the same endpoint. Trauma is the most direct: after a hemothorax, blood that is not fully evacuated organizes in the pleural space and the lung sits trapped within it. Tuberculous pleuritis produces the same result and remains a substantial cause of fibrothorax in high-burden countries. Inflammatory pleural disease reaches it through chronic pleural inflammation rather than through infection or blood. Renal disease was named as a fourth [Interview 1].

That list matches the published one closely enough to be worth stating. Mutsaers and colleagues identify organized haemorrhagic effusion, tuberculous effusion, empyema, and asbestos-related pleurisy as converging on the same fibrotic endpoint [41], and Tucker and Idell add collagen vascular disease [46]. A surgeon reasoning from his own cases and a review reasoning from mechanism arrive at substantially the same set. The final common pathway is the organized collagenous peel; the insult that initiates it is variable. This is what justifies scoping the need across benign pleural disease — infection, retained blood, or inflammatory pleuritis — rather than restricting it to empyema alone.

The surgeon also raised malignant pleural effusion, and immediately set it apart: those patients are managed differently [Interview 1]. That distinction is the reason malignancy is excluded from this need statement. Malignant trapped lung is managed palliatively with indwelling pleural catheters, which control symptoms effectively and are not surgery. Including it would enlarge the population while destroying the central claim of this assessment — that for the target population, an operation is currently the only definitive option. The exclusion was made on scoping grounds before this interview; the surgeon's response corroborates it independently.

Two consequences follow. The benign etiologies beyond empyema — retained hemothorax, inflammatory pleuritis, renal disease — expand the addressable population beyond the parapneumonic funnel modelled in Section 5, and that expansion is not currently quantified anywhere in this report. Tuberculosis is different: it expands the clinical need considerably but not the US market as modelled, because the tuberculous fibrothorax burden falls disproportionately outside the United States. It belongs in a later geographic expansion case, not folded into the primary forecast.

Finding 1.4 — Procedure Volume

The surgeon estimated performing at least one decortication per week and characterized that as frequent relative to the rest of his operative practice [Interview 1]. At face value that is on the order of fifty procedures per year for a single surgeon. This should be handled as a statement about his practice, not as an incidence estimate — a thoracic surgeon observes the failure tail of pleural disease and cannot supply a denominator. But it is worth recording, because it sits badly against the secondary data. The STS General Thoracic Surgery Database captured 7,316 decortications across 263 institutions over 2009–2016 [3], averaging roughly three to four per institution per year even before accounting for the fact that not all institutions participated throughout. One surgeon reporting fifty is an order of magnitude above the per-institution rate.

Three explanations are possible and this report cannot distinguish between them. His institution may be a high-volume referral centre, which would make him a poor guide to the national picture. His use of "decortication" may be broader than the coded procedure the STS captures, taking in partial decortications and pleural débridements recorded differently. The STS-derived figure may substantially undercount national volume, since the database reflects only participating institutions and only certain coded procedures. Or the case mix at his site may simply differ from the national one.

The third explanation has direct support from the interview. The surgeon noted that his institution serves a patient population with substantial injection drug use [Interview 1]. Injection drug use is an established risk factor for pleural infection — through bacteraemia and septic pulmonary emboli, through aspiration, and through delayed presentation — so a centre with that case mix would be expected to run above the national average for empyema, and therefore for decortication. This is the most parsimonious explanation for the discrepancy, and it argues against generalizing fifty procedures per year to a national denominator.

Interview 2 — Cardiothoracic ICU Physician Assistant

CTICU PA; personally instills intrapleural enzymatic therapy. Interviewed as the executor of the current standard of care. Site volume approximately 2–5 empyema or complicated effusion cases per month, representing a failure-enriched subset of the institution's total.

Finding 2.1 — The original need statement was wrong

This project began with the hypothesis that the labor burden of the enzymatic instillation protocol — repeated dosing, clamping, unclamping, nursing time — constituted the unmet need. That hypothesis did not survive contact with the person doing the work.

The PA, not the bedside nurse, performs the instillation. Asked directly whether the burden was material, the answer was that one patient on the protocol is 'not that disruptive,' and that three or four simultaneously becomes 'cumbersome' but occurs rarely enough to absorb. Presented with the workflow-automation value proposition, the PA declined it.

A prospective customer declining to buy is the cleanest possible kill signal, and the workflow need was retired on that basis. It is documented here rather than quietly dropped because the discipline of killing a need on primary evidence is the methodology working correctly.

Finding 2.2 — The protocol as actually executed

tPA and DNase are delivered as two 60cc pharmacy-prepared syringes, dosed every twelve hours for six total doses, with the chest tube clamped for a one-hour dwell before reopening. Missed doses occur approximately 5% of the time, concentrated overnight, with no observed clinical consequence — a detail that quietly undercuts the premise that dosing precision is where value lives.

Finding 2.3 — Bimodal response

Enzymatic therapy 'worked really well or didn't work at all,' and the outcome is frequently evident after the first dose. Slightly more than half of cases resolve without surgery at this site. This bimodality is not explained in the literature and is worth pursuing: if a first-dose response reliably predicts the final outcome, the six-dose protocol is exposing non-responders to three days of delay against a five-day surgical timing threshold [3].

Finding 2.4 — The real workflow hazards

The pain points the PA volunteered were not about labor. They were about failure modes with patient consequences. First, remembering that the tube must not be left clamped past the dwell — prolonged clamping risks hemothorax, and the PA has witnessed three-liter bleeds. Second, a limitation in tube maintenance: a clogged chest tube cannot be cleared, because the agent used to clear a clogged tube is the same lytic that is already being given.

Finding 2.5 — Prediction is not the gap

An early candidate need was a tool to predict enzymatic-therapy failure. The PA's response reframed this: clinicians frequently already know which patients will fail, and run the full course anyway — because the patient is a poor surgical candidate, because the patient refuses surgery, or because the team wants to exhaust non-operative options before committing. Prediction is not the bottleneck; the bottleneck is that knowing does not change what is available to do. The residual value of a predictive tool survives only in the subgroup that is genuinely operable, where earlier escalation would move a patient inside the five-day window [3].

This is worth stating plainly because it cuts against the published consensus. The field's own leaders identify personalized therapy and bedside dosing guidance as the priority development opportunity in intrapleural fibrinolytic therapy [39], and a review of the area frames the absence of a predictive biomarker as the central gap [31]. The primary data collected here suggests that framing is incomplete: at bedside, the binding constraint is not that clinicians cannot predict failure, but that prediction does not change what they are able to offer. A tool that tells a team their frail eighty-year-old will fail enzymatic therapy does not produce a treatment for that patient.

Procedure Observation — Decortication

Direct observation of a decortication provided the sensory basis for the mechanical argument in this report: the physical effort of stripping the peel, the visible fragility of the lung surface beneath it, and the extent to which the operation is a manual tissue-separation task rather than a precision-instrument task.

The case was performed thoracoscopically and completed in approximately ninety minutes without conversion to open thoracotomy. It was not a single procedure but a composite of three distinct acts: drainage of the pleural collection, decortication of the organized peel from the visceral pleura, and parietal pleurectomy — stripping the pleural lining from the chest wall to obliterate the pleural space and promote adherence of the re-expanded lung against it.

That composite structure bears directly on Finding 1.4. The question of whether a surgeon's use of the word decortication corresponds to the coded procedure captured in the STS database is not hypothetical: the case observed here spans several distinct CPT descriptors — drainage, decortication, and parietal pleurectomy are separately defined [43] — and how such a case is recorded determines whether it surfaces in a national procedure count as one decortication, as a pleurectomy, or as something else again.

Crucially, the surgeon characterized this case as one of the simpler ones, likening the dissection to "peeling an orange" [Interview 1]. The comparison is more exact than it is casual: peeling describes the separation of an adherent layer from a fragile surface by hand and by traction, where the failure mode is tearing into what lies beneath. Harder cases run two to three hours, and the reason he gave is the one this report is built on — the rind has been organizing for longer and has hardened. Operative duration is therefore not a fixed property of the procedure but a function of substrate maturity, the same variable that determines whether enzymatic therapy works at all [30].

That relationship matters for the economic model in Section 5. At roughly \$26 per operating-room minute [14], the ninety-minute case observed here represents about \$2,300 in operating-room time, while a three-hour case represents about \$4,700. The value of any tool that shortens dissection is therefore concentrated in exactly the cases where the need is most acute, and an average-case economic model understates it. This mirrors the segmentation applied to the adjacent adhesiolysis market in Section 5: the commercially relevant population is not every case but the difficult subset, where existing instruments struggle and operative time and cost run highest.

It also reopens a timing argument on ground that MIST-3 does not occupy. The randomized evidence discussed in Section 3 bounds the value of moving faster within the first days of treatment, measured against length of stay [37]. The surgeon's observation concerns a different timescale and a different endpoint: weeks of continued organization producing an operation that takes twice as long. Those are separate claims, and the second is untested. If delay to surgery makes the operation materially harder rather than merely later, that is a cost the length-of-stay literature would not detect.

What the Primary Research Changed

Hypothesis going in	What primary data showed	Status
Nursing labor burden of lytic protocol is the need	PA performs instillation, declined the value proposition	KILLED [Interview 2]
Predicting lytic failure is the need	Clinicians already know; candidacy blocks acting on it	NARROWED to operable subgroup [Interview 2]

Hypothesis going in	What primary data showed	Status
Reaching loculated pockets is the need	Failure is substrate, not access	REFRAMED [Interview 1, 2; 30]
Organized rind is the need	Corroborated independently by surgeon and by ML predictor data	LOCKED [Interview 1; 30]
Rind removal is mechanically traumatic	Confirmed — blunt force stripping with forceps	NEW — became device-route premise [Interview 1]

5. Market Opportunity

Top-Down Sizing

The honest top-down statement is a range rather than a point. Anchoring on the ~90,000 serious US pleural infections per year cited in the LTI-01 literature [9] and the ~40,000 requiring pleural drainage [11], and applying the 20–30% surgical-drainage fraction [2, 7], the surgical population falls somewhere between 8,000 and 27,000 procedures annually. The STS captured 7,316 decortications over roughly seven years across 263 participating institutions [3], which is consistent with a national figure in the lower half of that band once non-participating hospitals are added back.

The ~10,000 figure is built by multiplying through the parapneumonic funnel, and it is worth showing the arithmetic explicitly because the compounding uncertainty is large.

The AATS consensus guidelines put US pneumonia hospitalizations at approximately one million annually, of which 20–40% develop a parapneumonic effusion and 5–10% of those effusions progress to empyema [2]. Taking the low end of each band gives 10,000 empyemas per year; taking the high end gives 40,000. The literature therefore supports a fourfold range before the surgical fraction is applied at all.

The surgical fraction is itself contested. The AATS guidelines state that approximately 30% of empyema patients require surgical drainage [2]. A systematic review of 134 articles covering 227,898 patients puts it at approximately 20%, with an interquartile range of 1–32% [7]. That IQR is the more informative number: it indicates that across the published literature, the proportion of empyema patients taken to surgery varies enormously by institution, era, and threshold for escalation. These are not two estimates of one quantity; they are a description of practice heterogeneity.

Applying 20–30% to the 10,000–40,000 empyema band yields a decortication range of 2,000 to 12,000 per year. Compounding four multiplicative uncertainties produces a sixfold spread, which is what a funnel of this kind should be expected to produce.

. Tucker and colleagues estimate approximately 40,000 US patients require pleural drainage annually [11]; applying the same 20–30% surgical fraction gives 8,000–12,000, a band that brackets the working figure closely. Against this, the STS General Thoracic Surgery Database captured 7,316 decortications across 263 institutions over 2009–2016 [3] — roughly 915 per year among participating centres. Converting that to a national figure requires knowing what fraction of US thoracic surgical volume the database captures, which this report has not established, so the STS number constrains the estimate only weakly.

The ~10,000 used here sits at the upper end of the funnel-derived range and the upper end of the drainage-derived range. It is not a midpoint and should not be presented as a conservative assumption — on the literature alone, a figure closer to 6,000 would be equally defensible. Two considerations argue against revising down: the surgeon interviewed for this project reports roughly fifty decortications a year on his own (Finding 1.4), and the composite coding structure observed in a single case suggests decortication volume may be systematically under-captured by procedure-code counts. The honest summary is that the true figure is unknown within roughly a factor of two in either direction, and that every revenue figure in this section inherits that uncertainty.

One further inconsistency should be noted rather than smoothed over. The LTI-01 Phase 1 publication cites approximately 90,000 serious pleural infections annually in the US [9], which exceeds the upper bound of the empyema band derived above. The likely explanation is definitional — "serious pleural infection" plausibly includes complicated parapneumonic effusions that would not be classified as empyema.

Growth

Pleural infection hospitalizations rose 37.5% over 2007–2016 [1], and empyema incidence has been increasing since the 1990s [2]. A 6.3% annual growth assumption is used in the forecast below, which sits below the historical rate implied by [1] and is therefore conservative.

Bottom-Up Forecast — Device Route

Price for the device route is derived from the value drivers it acts on, each capped by the share of value a purchaser will concede. Because most of the levers avert probabilistic events rather than displacing a hard cost, the indirect capture band of 10-30% applies to them; operating-room time is treated separately at 30-40%, being a direct and immediately realizable saving.

Value driver	Derivation	Value per case
OR time saved	\$26/min [14] — 10-30 min saved	\$260-780 (or \$500-1,000 if freed slots convert to added throughput)
Conversion avoided	10-25% relative reduction — \$3,476 gap [15]	\$347-869
Prolonged air leak (PAL) avoided	7.5% rate [5] — 50% cut — \$2,556 [16]	~\$96
Length of stay	1 fewer day — ~\$2,000-2,500, less ~25% overlap with PAL	\$1,500-1,900

Applying 10-30% capture to the indirect lever and 30-40% to operating room time gives an implied ASP of \$450-650. That range is independently proved against observed purchasing behavior: fibrin sealant is bought into these same cases at roughly \$480 per application [17], establishing that the price point sits below demonstrated willingness to pay rather than testing it.

Forecast assumptions: ASP \$550; 10,000 base procedures growing 6% annually; one unit per procedure; S-curve adoption from 3% to 30% following standard diffusion-of-innovation adopter segmentation [18], with innovators and early adopters concentrated in high-volume academic thoracic programmes.

Year	Addressable procedures	Penetration	Units	Revenue at \$500 ASP
1	10,000	3%	300	\$170,000
2	10,600	8%	848	\$470,000
3	11,236	15%	1,685	\$930,000
4	11,910	22%	2,620	\$1.44M
5	12,625	30%	3,788	\$2.1M
7	14,185	45%	6,383	\$3.51M

The device route's US decortication-only market is small — roughly \$5–6M at full penetration, with 75% gross margin achievable if COGS is held under ~\$140.

Adjacent Market — Adhesiolysis

The mechanical task in decortication is atraumatic separation of an adherent fibrous plane from lung surface. That same task appears in a far larger population: lung cancer resection in patients with pleural adhesions. Approximately 56,000 US lung cancer resections occur annually [20]. Adhesions are present in roughly 35–41% of these cases — 34% of 144 major lung resections in one peer-reviewed cohort [21], and 41.4%

across 952 graded lobectomies in a second series [22]. Critically for market sizing, only the severe or dense subset requires a dedicated separation tool: approximately 10–12% of patients, corresponding to the 112 of 952 patients with adhesions involving more than one lobe [22]. Adhesions independently predict conversion to thoracotomy and postoperative surgical complications [19].

The addressable adjacent market is therefore the tool-requiring subset, not all adhesions. Applying 10–12% to the 56,000-procedure base yields approximately 5,600–6,700 additional annual cases. At \$550 ASP that is \$3.1–3.7M, bringing the combined decortication-plus-adhesiolysis opportunity to roughly \$8.1–8.8M at full US penetration. Sizing on the broader 35–41% 'any adhesions' band would yield 19,600–23,000 cases and \$10.8–12.7M, but that band includes filmy adhesions readily divided with existing instruments and overstates the true opportunity. Decortication is the beachhead because the need is most acute and the comparator is worst; dense adhesiolysis is the volume expansion.

Bottom-Up Forecast — Biologic Route

The biologic population is derived from the pleural drainage denominator rather than the pneumonia funnel. Tucker and colleagues estimate approximately 40,000 US patients require pleural drainage annually [11]. Applying a 50% enzymatic failure rate gives 20,000 patients per year who fail intrapleural enzyme therapy and constitute the rescue population.

Of those, a therapy that reverses the organized rind is assumed to prevent surgery in 40%, yielding 8,000 decortications avoided annually. At approximately \$20,000 per surgical episode [33], total annual value created is 8,000 multiplied by \$20,000, or \$160M. Applying a 60% shared-ROI capture — the conservative end of the 60-80% band the framework permits for direct cost displacement — gives a serviceable addressable market of \$96M.

Price then follows from total value multiplied by capture and divided by number of uses. The agent is administered to all 20,000 rescue patients, not only the 8,000 in whom surgery is ultimately averted: \$96M divided by 20,000 uses gives an ASP of \$4,800 per course. The distinction matters and should be stated when the number is challenged — value is concentrated in 40% of treated patients while price is spread across all of them.

That price clears the payment environment it would enter. Empyema admissions are reimbursed at roughly \$7,000-11,000 on a medical DRG against \$17,000-31,000 on a surgical DRG [CMS FY2025 IPPS Final Rule, MS-DRG relative weights — see Appendix E]. A five-dose tPA/DNase course at approximately \$15,000 [32] already consumes most of the medical DRG on its own. Preventing an operation moves the patient off the surgical DRG entirely, and \$4,800 sits below both the operation it averts and the enzymatic course it rescues.

Assumptions: ASP \$4,800; addressable population defined as patients failing or predicted to fail enzymatic therapy, estimated at ~20,000 annually growing 6.3% per year; single treatment course per patient. Revenue begins only after a 7–10 year development period.

Year post-launch	Addressable patients	Penetration	Revenue at \$4,800 ASP
1	20,000	4%	\$3.84M
2	21,260	9%	\$9.18M
3	22,600	16%	\$17.4M
5	25,540	30%	\$36.7M
7	28,860	42%	\$58.2M

At a pharmaceutical industry weighted-average COGS of 25.7% [34], a \$4,800 ASP implies roughly \$1,234 in cost of goods and a 74% gross margin — before the amortized development cost, which is the dominant economic factor in this route.

Purchaser Willingness-to-Pay

Cost of the status quo

Cost element	Figure	Source and caveat
Empyema admission, mean total	\$38,591	National analysis [1]
Mean length of stay	13.8 days	National analysis [1]
Hospitalized >1 month	~25%	MIST2 population [29]
tPA/DNase, 5-dose course	~\$15,000	Kheir et al. [32]. Single institution.
Medical thoracoscopy	~\$9,700	Kheir et al. [32], same comparison
VATS	~\$20,000	Lacin et al. [33]
VATS-to-open 90-day cost gap	\$3,476	Farjah et al., MarketScan, 9,962 patients [15]. Lobectomy population; average approach gap, not measured conversion cost.
Prolonged air leak, added cost	~\$2,556 (range \$571–3,573)	Jones et al., pooled systematic review [16]. Lobectomy data, not decortication.
OR time	~\$26/minute	Grossi et al., Premier multi-institution [14]. Valve-surgery denominator; thoracic-adjacent proxy only.

Value capture logic

For the device route, the value proposition is avoidance of costs the hospital already bears: conversions to open, prolonged air leak, operating-room minutes, and inpatient days. The full driver-by-driver derivation and its capture assumptions are set out above; the summary position is that roughly \$2,900 of value is created per average case and \$550 of it is captured, a capture rate of 15-22% that sits inside the 10-30% band the framework permits for indirect savings.

One mechanistic detail from the Farjah analysis strengthens this argument and should be used explicitly. The \$3,476 gap between VATS and open was largely explained by differential rates of prolonged length of stay; once adjusted for prolonged stay, the residual difference fell to \$1,276 and lost significance [15]. In other words, the approach itself is not what costs money — the complications that follow the harder operation are. That is precisely the causal chain a device claiming reduced conversion and reduced air leak would interrupt, and it means the economic argument should be framed around avoided prolonged stay rather than around approach substitution.

For the biologic route the logic is stronger and simpler: a \$4,800 course priced against a \$15,000 enzymatic course [32] and a ~\$20,000 VATS episode [33] that it would avert, within a \$38,591 admission [1]. Indirect and avoided-event savings typically support capturing 10–30% of value; here the direct substitution economics support more.

6. Stakeholder and Adoption Landscape

Job-to-be-Done

The core functional job is: restore and preserve re-expansion of the lung against the chest wall in a patient whose pleural space has become occupied by infection, blood, or inflammatory exudate.

Job step	What the executor is trying to do	Representative desired outcomes
1. Define	Establish that the pleural space is the problem and determine what stage it has reached	Minimize the time it takes to distinguish a simple effusion from a complicated or organized one; minimize the likelihood of misclassifying stage
2. Locate	Identify the collection, its loculations, and a safe access route	Minimize the time it takes to localize all fluid compartments; minimize the likelihood of failing to identify a compartment that will not drain
3. Prepare	Establish drainage access and ready the therapy	Minimize the time it takes to establish adequate drainage; minimize the likelihood of drain malposition
4. Confirm	Verify the drain is functioning and determine which treatment paths the patient is eligible for	Minimize the likelihood of proceeding with a non-functional drain; minimize the time it takes to determine surgical candidacy
5. Execute	Clear the pleural space and free the lung so it can expand	Minimize the time it takes to clear the space; minimize the likelihood of injuring the lung while doing so; minimize the invasiveness required
6. Monitor	Determine whether the chosen treatment is working	Minimize the time it takes to know whether the treatment will succeed; minimize the likelihood of continuing a futile course
7. Modify	Escalate, de-escalate, or change approach when the current path is failing	Minimize the time it takes to escalate once failure is apparent; minimize the likelihood of escalating after the patient has deteriorated
8. Conclude	Confirm durable re-expansion and return the patient to baseline function	Minimize the likelihood of recurrence or persistent entrapment; minimize length of stay; minimize residual loss of pulmonary function

Where satisfaction is lowest

Steps 1 through 3 are well served. Imaging characterizes pleural collections adequately, image-guided drainage is routine, and the interview site reports the mechanics of drain placement and enzymatic installation as unremarkable [Interview 2]. Step 8 is well served when the preceding steps succeed: decortication restores measurable pulmonary function, with FVC improving from roughly 70% to 85% of predicted in one series [6].

Three steps are underserved:

Step 5, Execute — the central gap. There is no non-surgical means of clearing an organized rind. Fibrinolytics act on fibrin and fail once the substrate is collagen, which is why pleural thickening dominates failure prediction [30]. The only method that works applies tearing force to a peel adherent to the visceral pleura, which is the same force that injures the lung, at a cost of 7.5% prolonged air leak [5] and 39.3% overall complications [3]. The desired outcomes 'minimize the likelihood of injuring the lung' and 'minimize the invasiveness required' are both badly unmet, and they are unmet simultaneously because the only available mechanism trades one against the other.

Step 6, Monitor — unmet but not decisive. No biochemical test or imaging modality can assess a patient's fibrinolytic system or predict response to enzymatic therapy [31], and the optimal dose and dosing interval remain unresolved after seventy years [39]. The interview site observes response to be bimodal and frequently evident after the first dose [Interview 2], which suggests the information exists earlier than the protocol acts on it.

Step 7, Modify — unmet for a non-technical reason. This is where primary research changed the analysis. Clinicians frequently know a patient will fail and complete the course regardless, because the patient is a poor surgical candidate, refuses surgery, or the team wants to exhaust non-operative options [Interview 2]. The constraint on Step 7 is therefore not information but the absence of an option to escalate to. Improving Step 6 does not relieve it. Half the patients allocated to surgery in a trial designed to eliminate fitness-based selection did not receive it [37].

The strategic conclusion follows directly from the job map. Steps 6 and 7 are gated by Step 5: better prediction and faster escalation are only valuable if there is something to escalate to. A solution addressing Step 5 relieves all three; a solution addressing Step 6 alone relieves none of them for the population that cannot have an operation. This is the analytical basis for locking the need statement onto the substrate rather than onto workflow or prediction.

Job Executors

Stakeholder	Functional job	Emotional / social job	Current dissatisfaction
Thoracic surgeon	Remove the rind and restore re-expansion without injuring lung	Be the definitive answer; avoid preventable air leak and reoperation	Moderate — the operation works; the frustration is the patients who cannot have it [Interview 1]
Interventional pulmonologist	Resolve the pleural space without handing the patient to surgery	Own the definitive intervention	High — no tool for organized disease [31]
CTICU PA / intensivist	Execute the protocol safely; recognize failure early	Not be the person who left the tube clamped	Moderate — hazards, not labor [Interview 2]
Hospital / payer	Discharge alive, on time, without readmission	Avoid outlier cost against a fixed DRG	High — 13.8-day mean LOS, 25% over a month [1, 29]
Patient	Breathe normally again	Avoid an operation; avoid chronic pain	High — up to 44% chronic pain post-thoracotomy [28]

Adoption Enablers

The clearest enabler is the existence of a population with no current option. A surgeon has no incentive to replace an operation he performs well, but he has every incentive to acquire an answer for the patient he must decline. The value proposition that resonated in primary research was framed exactly this way — that if the rind could be eliminated without surgery, nobody would miss doing decortications [Interview 1].

There is also randomized evidence that patients fare better on non-surgical management by the measures they themselves report. In MIST-3, health utility at two months was significantly higher after enzymatic therapy than after VATS (EQ-5D index 0.83 versus 0.59, $P = 0.023$), and mean pain score at two months was 4.9 versus 22.2 on a 100-mm scale [37]. The trial was small and not powered for these endpoints, and inpatient pain was actually highest in the enzymatic arm, so the effect is a signal rather than a finding. But its direction is consistent with the chronic post-thoracic-surgery pain literature [28] and supports a premise this business case depends on: a non-surgical route to re-expansion would be actively preferred by patients, not merely tolerated.

Secondary enablers: the pleural space is directly accessible through hardware already in the patient, which removes a delivery barrier that would sink most locally-acting therapies; the outcome is objectively measurable on imaging within days, which supports both a clean trial endpoint and a fast institutional evaluation; and the comparator's morbidity is high enough [3, 5, 28] that a moderately effective alternative clears the bar.

Adoption Barriers

Five barriers stand between either route and adoption. The first is practice-pattern inertia: empyema management has historically varied substantially between institutions and between specialties [2], which cuts both ways — there is no single entrenched standard to displace, but equally no single decision-maker to convince. The second is reimbursement structure, since both routes are consumed inside an inpatient stay paid by a fixed DRG. A device is bundled into the surgical case and represents pure added cost unless offset is demonstrated; a biologic used in the inpatient setting is likewise bundled, and separate payment would require either NTAP as a three-year bridge or migration of the procedure to an outpatient setting where a J-code can be separately paid. Related to both is the question of specialty ownership: the escalation decision sits between pulmonology and thoracic surgery and is genuinely contested, driven by patient age and surgical risk, so a product arriving without a clear owner risks being nobody's decision. Fourth is evidence burden — the endpoint that matters clinically, avoided surgery, requires following patients past the acute admission, which lengthens any trial and raises its cost. Fifth is competitive timing: LTI-01 has completed Phase 2a and is planning Phase 2b [10], and if that programme broadens into organizing disease, first-mover advantage in the biologic route erodes.

7. Technology and Feasibility

The Mechanistic Constraint

Any solution has to contend with one fact: the mature rind is cross-linked collagen, not fibrin. This is why the existing drug class cannot be dose-escalated into effectiveness, and it eliminates several intuitively appealing alternatives.

- Fibrinolytics — tPA, DNase, urokinase, single-chain urokinase — cleave fibrin and degrade DNA. They do not degrade collagen. Higher doses do not change the substrate mismatch, and the LTI-01 Phase 2a data showed the highest dose tested (1,200,000 U) performing worst on pleural opacification [10].
- Anti-fibrotics — nintedanib, pirfenidone — attenuate the formation of new fibrosis. They do not dissolve collagen that has already been deposited. Proposing them for established rind is a mechanism error.
- Collagenase does degrade Type I and III collagen and is therefore mechanistically correct — but no product has ever been approved for pleural use, which forecloses the 505(b)(2) shortcut.
- Blocking or reversing mesomesenchymal transition is an active research direction with named molecular targets, pursued most visibly by the UT Tyler group [46]. It is mechanistically distinct from the generic anti-fibrotics above, because it targets the specific cellular transition that produces pleural myofibroblasts rather than fibrosis in general. But the therapeutic goal stated in that work is blocking or reversing the differentiation of the cells, not degrading matrix those cells have already deposited. It is therefore a Stage II prevention strategy, not a Stage III reversal strategy, and does not on its own address an established rind.
- Mechanical separation works and is what decortication already does. The engineering question is whether the separation force can be applied more selectively than a surgeon's forceps.

Device Route — Water-Jet Hydrodissection

Concept. A pressure-controlled fluid jet exploits the difference in tensile strength between the collagenous peel and the underlying visceral pleura, separating the plane hydraulically rather than by blunt traction. The clinical target is the injury mechanism identified in Interview 1: replacing tearing force with a selective, tunable separation force.

Feasibility. High. Water-jet dissection is an established surgical mechanism with existing cleared devices in other tissue applications. The physics are well characterized and the adaptation required is geometric — pleural-appropriate tip design and working length — rather than fundamental.

Economics. Target ASP \$450–650 with COGS under \$140 yields ~75% gross margin. The reference price point is validated by fibrin sealant purchasing at ~\$480 per application in thoracic surgery [17].

Validation path. Ex vivo decorticated rind specimens for plane-separation testing against blunt dissection as comparator; measurement endpoints are separation force, visceral pleural perforation rate, and time to clear a standard surface area. Then a rabbit chronic-empyema model, for which validated protocols exist in the pleural fibrinolytic literature [40].

IP risk. This is the dominant risk and it is significant. Building on an existing cleared water-jet mechanism means either designing around held patents or licensing them. This should be framed in all materials as a feasibility reference demonstrating that the physics work in surgical tissue — not as a committed solution architecture. A freedom-to-operate analysis is the essential first step in Section 9.

Biologic Route — Intrapleural Anti-Collagen Therapy

Concept. Intrapleural delivery of a collagen-degrading enzyme to dissolve the organized peel, restoring re-expansion without an operation. Delivered through the chest tube already in place.

Feasibility. Moderate on mechanism, low on delivery. The enzymatic chemistry is sound. The unsolved problem is physical: a mature rind is dense and poorly perfused, and an agent instilled into the pleural space will contact its surface rather than penetrate its thickness. Whether a clinically meaningful depth of degradation can be achieved without exposing the underlying lung to the same enzyme is the central unanswered question, and it is the one that a Phase 1 would have to resolve.

Selectivity risk. Collagen is not unique to the rind. The visceral pleura and lung parenchyma contain collagen. A non-selective collagenase risks creating exactly the injury the device route is designed to avoid, in a less controlled way. This risk should be stated openly in any pitch — it is the first question a thoracic surgeon will ask.

Precedent for pleural biologic development. The LTI-01 program establishes that intrapleural biologics can be developed and safely administered: no bleeding events at any dose across Phase 1 (n=14) [9] and Phase 2a (n=40) [10]. That is a meaningful de-risking precedent for the route even though the molecule is mechanistically different.

Regulatory reality. Full BLA. No pleural collagenase predicate exists, so 505(b)(2) is unavailable. Seven to ten years, high attrition.

Competitive Landscape — LTI-01

LTI-01 is a recombinant single-chain urokinase plasminogen activator developed by Lung Therapeutics, Inc., subsequently acquired by Aileron and now Rein Therapeutics; the underlying patent is held by the University of Texas Board of Regents and licensed to LTI [10]. It is the direct competitor in the sense that it targets the same patients at the same decision point.

Phase 1 (n=14, open-label dose escalation, 50,000–800,000 IU daily up to three days) established tolerability with no bleeding, no treatment-emergent adverse events, and no surgical referrals, with reduced pleural opacity in all but one subject [9].

Phase 2a (LTI-01-2001, NCT04159831) was a randomized, double-blind, placebo-controlled, multicenter dose-ranging trial at 18 US sites. Target enrollment was 100–160; COVID-19 limited it to 43 enrolled and 40 dosed. The primary endpoint — treatment failure within 7 days — was not met: 37.9% failure across all LTI-01 arms versus 36% placebo. Two prespecified sensitivity analyses showed favorable trends at the 400,000 U dose (p=0.052 and p=0.147). Pleural opacity volume improved significantly versus placebo in absolute terms at 400,000 U and 800,000 U (p<0.05), with the 800,000 U group showing a median relative reduction of 55.8% versus 16.1% for placebo. Incidence of surgery at day 30 was 20.7% across LTI-01 arms versus 45.5% for placebo, not statistically significant. No bleeding events occurred at any dose [10].

Why this is strategically favorable, not threatening

Three specific features of the Phase 2a report support rather than undermine this project's need statement.

First, the trial explicitly excluded patients with fibrothorax and with non-expandable lung preceding the infectious episode [10]. The organized-rind population was designed out of the study.

Second, and more pointedly, the investigators reported that situations in which subjects required surgery for evolving pleural fibrosis or trapped lung were not well captured by the protocol's definition of treatment failure — and that in three patients, rescue surgery was performed despite the protocol scoring them as treatment successes [10]. The competitor's own trial design failed to see the failure mode this project is built around.

Third, the stated conclusion is that a larger Phase 2b is required to confirm these results or to determine efficacy in patients with organizing, non-draining, infected pleural effusions [10]. Efficacy in organizing disease is, by the sponsor's own account, unestablished.

The strategic read: the leading pleural biologic in development is a fibrinolytic that missed its primary endpoint in a COVID-truncated trial, that excluded organized disease by protocol, and whose investigators identified trapped lung as a category their endpoints could not measure. This is a competitor validating that the space exists rather than occupying it.

Intellectual Property Position

Route	White space	Exposure	Assessment
Device	Pleural-specific tip geometry, pressure profiles tuned to rind vs. visceral pleura, integrated evacuation	Existing water-jet hydrodissection patents in surgical tissue applications	WEAK — design-around or license required; FTO analysis is gating
Biologic	Pleural-indication collagen degradation; formulation and penetration-enhancing delivery	No pleural collagenase predicate found; LTI-01 patent is fibrinolytic and mechanistically distinct	MODERATE — genuine white space, but patentability depends on enzyme sourcing

Named Analytical Risk: Confirmation Bias

An inherent risk in this assessment is the potential to reason backward from a preferred solution to a justifying need. To mitigate this, specific controls were implemented: the need statement was locked prior to solution development; the initial workflow need was discarded following negative customer feedback rather than reframed; the predictive-failure need was narrowed when primary data suggested otherwise; and the mechanistic constraint section was structured to eliminate solution classes rather than justify specific ones.

The remaining exposure lies in the fact that both surviving routes were generated from the same premise—that rind must be removed or dissolved—without serious pursuit of alternative framing. Unexplored options include pharmacological prevention of rind formation during Stage II, restaging protocols that escalate earlier based on first-dose non-response signals [Interview 2], and accepting entrapment while optimizing the physiology of the remaining lung.

8. Regulatory and Distribution Considerations

Device Route — Regulatory Fork

Option A — 510(k), tool claim. Claim substantial equivalence to a cleared hydrodissection or surgical dissection device; indication written as tissue dissection in the pleural space. Estimated 6–12 months, bench and animal data, no clinical trial. Fast and cheap. The cost is that a tool claim supports no economic argument — a device that merely dissects is bundled into the DRG and has no basis for separate payment or premium pricing.

Option B — de novo, outcome claim. Claim reduction in air leak or reduction in conversion to thoracotomy. This is the claim that generates the health-economic argument in Section 5 and justifies the price. It requires clinical data and a longer, costlier pathway — roughly 12–24 months beyond bench work.

Recommendation: Launch the product using the faster, simpler regulatory path (Option A) to begin generating revenue immediately. In parallel, continue collecting performance data through registries and independent studies. This real-world evidence can support an expanded, more valuable claim later, allowing the project to prioritize early cash flow rather than delaying market entry for expensive clinical trials.

Biologic Route — Regulatory Path

Full BLA. IND-enabling toxicology, then Phase 1 safety and proof of intrapleural mechanism, Phase 2 dose-ranging and efficacy, Phase 3 pivotal, then BLA review. Seven to ten years to approval.

Two design lessons are directly transferable from the LTI-01 Phase 2a experience and should be adopted rather than rediscovered [10]. First, the definition of treatment failure must explicitly capture surgery for evolving fibrosis or trapped lung — the competitor's protocol did not, and it produced patients scored as successes who went to the operating room. Checking CT scans to measure lung cloudiness is a reliable way to see if a treatment is working. Patients whose lung cloudiness barely decreased had a 90% chance of treatment failure, compared to only 20% for those who showed significant improvement. Because of this, using CT scan measurements of cloudiness is a proven tool that can be used to design future clinical trials.

Reimbursement Pathways

Element	Device route	Biologic route
Existing codes	Procedure already coded: 32220, 32225, 32320 (open); 32651, 32652 (VATS) [43]	None — new NDC to J-code required
Inpatient payment	Bundled into DRG	Bundled into DRG
Bridge mechanism	None typically needed at this ASP	NTAP, 3-year bridge, then DRG recalibration
Physician payment	Included in existing surgical CPT	New CPT, beginning Category III
Path to separate payment	Requires outcome claim plus health-economic evidence	Outpatient administration would permit separate drug payment

Distribution and Commercial Model

Device. Direct hospital sales into thoracic surgery through value analysis committees, with GPO contracting as the scale channel. Single-use disposable, no capital equipment, which removes the largest adoption barrier. Clinical support is a half-day in-service rather than a training program. The realistic end game is not a standalone company — it is a product improvement built to sell or license to an established thoracic strategic, which is the lowest-capital path in a competitive market.

Biologic. Specialty pharmaceutical distribution through hospital pharmacy. Requires medical affairs, KOL development, and a field medical team. Not a viable independent path at student-project or seed scale; the realistic model is preclinical and Phase 1 value creation followed by partnership or licensure to a respiratory-focused pharmaceutical company.

9. Next Steps and Investment to Decision

Priority 1 — Close the Primary Research Gap

Pulmonology / interventional pulmonology interview. The single highest-value outstanding activity. Pulmonology holds the real denominator, owns the day-to-day judgment of whether enzymatic therapy is working, and drives the escalation decision that any new intervention must insert itself into. Specific questions: how many complicated effusions and empyemas the service manages annually versus how many reach surgery; what triggers a surgical consult and how contested that decision is; whether the bimodal first-dose response pattern [Interview 2] is recognized; what fraction of patients are formally declined for surgery and what happens to them; and whether a non-surgical rind therapy would be adopted by pulmonology or ceded to surgery. Cost: negligible. Timeline: 2–4 weeks.

Close the three open questions with the thoracic surgeon. First, the total annual decortication volume at the site across all surgeons, which converts his personal estimate into a figure comparable to the STS per-institution rate. Second, whether he counts partial decortication and pleural débridement as decortication; if so, the order-of-magnitude discrepancy in Finding 1.4 is largely definitional. Third, roughly what share of his decortications occur in patients with injection drug use — at a third or more this is a distinct clinical segment with its own surgical-candidacy profile rather than site colour. Cost: negligible. Timeline: 1–2 weeks.

Priority 2 — Replace Derived Numbers with Counted Ones

HCUP / NIS procedure-code pull. Query ICD-10-PCS codes for open and thoracoscopic decortication to produce a counted national annual volume, replacing the derived ~10,000 estimate that currently underpins every market figure in this report. Also stratify by patient age and comorbidity to size the non-surgical-candidate subpopulation, which is the segment with no current option and therefore the highest-value target. Cost: \$2–5K for data access and analyst time. Timeline: 4–8 weeks.

Priority 3 — Technical De-Risking

Freedom-to-operate analysis, device route. Patent landscape and FTO opinion on water-jet and hydrodissection surgical tissue-separation claims. This is gating — a negative result redirects the entire device strategy. Cost: \$25–50K for a competent firm. Timeline: 6–10 weeks.

Ex vivo bench testing. Decorticated rind specimens; measure separation force, visceral pleural perforation rate, and clearance time for hydrodissection against blunt instrument dissection. This is the experiment that either substantiates or refutes the central device claim. Cost: \$40–80K. Timeline: 3–4 months.

Animal model feasibility. Rabbit chronic-empyema model, for which validated protocols exist in the intrapleural fibrinolytic literature [40]. Cost: \$80–150K. Timeline: 4–6 months.

Biologic route — penetration study. Before any investment in formulation, establish whether an enzyme instilled into the pleural space penetrates a mature rind to clinically meaningful depth. Ex vivo, using human decortication specimens. This is a cheap experiment that answers the question most likely to kill the route. Cost: \$20–40K. Timeline: 2–3 months.

Priority 4 — Competitive and Regulatory Monitoring

- Track publication of the LTI-01 Phase 2a manuscript and the design of the planned Phase 2b [10], specifically whether organizing disease is included.
- FDA pre-submission meeting for the device route once bench data exists, to confirm the 510(k) versus de novo fork before committing to a claim.

Investment Summary to Pivot / Persevere Decision

The activities above divide cleanly into work executable within a master's program and work that requires external funding and institutional infrastructure.

Tier 1 — executable within the program

Activity	How it is executed at student scale	Cost
Complete primary research: pulmonology interview, surgeon follow-up, PA verification	Student time	\$0
National decortication volume	Free HCUP query tools first; then confirm whether the institution already holds an NIS license before purchasing data	\$0–400
Patent landscape, device route	Google Patents, USPTO Public Search, Espacenet, Lens.org; technology transfer office will typically run prior-art on a student invention at no charge	\$0
Ex vivo rind separation testing (device)	Discarded decortication specimens with departmental materials testing frame; measure separation force, visceral pleural perforation rate, and clearance time against blunt dissection	\$500–2,000
Ex vivo collagenase penetration and selectivity (biologic)	Same specimens. Incubate rind in bacterial collagenase (a standard catalogue reagent, milligram quantities) in buffer at 37 degrees; quantify depth of collagen degradation by trichrome or picrosirius histology and degradation products by hydroxyproline assay. Add a selectivity arm using abattoir porcine lung as parenchymal surrogate, run under identical conditions.	\$1,000–3,000
Benchtop proof-of-concept	Syringe or peristaltic pump, tubing, machined nozzles, university maker space	\$500–3,000
Regulatory input	FDA DICE answers questions at no cost; Q-Submission carries no filing fee; regulatory affairs faculty	\$0–500
TIER 1 TOTAL		\$2,000–8,900

The biologic experiment is worth designing carefully because it can answer the question most likely to close the route, and it can answer it for a few thousand dollars rather than a few million. Section 7 identifies two unresolved problems with intrapleural anti-collagen therapy: whether an instilled enzyme penetrates a dense rind to clinically meaningful depth, and whether it does so selectively rather than degrading the collagen in lung and visceral pleura as well. Penetration is testable on human rind specimens directly. Selectivity requires paired parenchymal tissue, which decortication specimens do not supply — the peel is removed and the lung stays in the patient — so abattoir porcine lung is the practical surrogate, at negligible cost and with no ethical approval burden. A result showing that collagenase degrades porcine parenchyma as readily as human rind would be a serious finding against the biologic route, obtained early and cheaply.

Tier 2 — requires external funding and institutional infrastructure

Activity	Cost	Why it sits outside program scope
Formal freedom-to-operate opinion	\$25–50K	A landscape search establishes whether to continue; only a firm-issued opinion is relied upon by investors or acquirers
Animal model, rabbit chronic empyema [40]	\$80–150K	IACUC approval, animal facility access, weeks of husbandry and a trained team; not compressible into two semesters
FDA pre-submission meeting	\$30–60K	Free to file, but requires a device description and bench data package that does not yet exist
Engineering prototype and design iteration	\$150–300K	Design controls, materials selection, biocompatibility testing
Pilot clinical study (Section 8, Phase 2)	\$350–600K	IRB, enrollment, monitoring, data management
TIER 2 TOTAL	\$635K–1.16M	

Funding paths for Tier 2. NSF I-Corps for customer discovery; NIH SBIR Phase I, which requires a formed entity; university innovation and translational seed grants; and regionally, the North Carolina Biotechnology Center's translational research programs.

The pivot/persevere decision is therefore reached in two stages rather than one. Tier 1, at \$1,000 to \$6,300, resolves whether the mechanical premise holds, whether the patent landscape permits a design, and whether the national procedure denominator supports the market case.

10. Appendices

Appendix A — Consolidated Evidence Table

Claim	Figure	Source	Confidence
US pneumonia hospitalizations/yr	~1,000,000	[2]	High
Parapneumonic effusion rate	20–40%	[2]	Moderate — wide range
Empyema rate of effusions	5–10%	[2]	Moderate
Serious pleural infections/yr US	~90,000	[9]	Low — secondhand
Require pleural drainage/yr US	~40,000	[11]	Moderate
Require surgery	~20% / ~30%	[7] / [2]	Low — sources disagree
Decortications, STS 2009–2016	7,316 / 263 sites	[3]	High — counted subset
Operative mortality	3.1%	[3]	High
Complication rate	39.3%	[3]	High
Major morbidity	15.6%	[3]	High
Transitional care at discharge	26.3%	[3]	High
30-day readmission	8.7%	[3]	High
Preop / postop LOS median	4 / 7 days	[3]	High
Mean LOS, empyema admission	13.8 days	[1]	High
Mean cost per admission	\$38,591	[1]	High
Rise in pleural infection admissions	37.5% (2007–2016)	[1]	High
1-year mortality	20% (35% frail)	[4]	High
Nonop vs op 1-yr mortality	32.3% vs 18.8%	[23]	High figure, biased comparison
Nonop vs op 30-day mortality	16.6% vs 5.4%	[24]	High figure, biased comparison
Real-world enzymatic success	78%	[30]	High — multicenter, 466 pts
Enzymatic failure rate	up to 25%	[38]	Moderate
Reported cure rate, unstratified	~90%	[28]	Low — no stage stratification
tPA/DNase bleeding rate	4.2% (n=1,850)	Akulian [36]	High — multicenter
Prolonged air leak post-decort	7.5%	[5]	High — on-population
Chronic pain, thoracotomy	up to 44%	[28]	Moderate

Claim	Figure	Source	Confidence
Persistent pain, VATS	~33%	[28]	Moderate
VATS conversion, Stage II vs III	8.4% vs 19.2%	[26]	High
VATS conversion, range across series	4.5% – 44.4%	[12], [13]	Wide — selection-dependent
FVC recovery post-decortication	69.74% → 85.40%	[6]	Moderate — single series
Hospitalized >1 month	~25%	[29]	High
Standard care failure rate	33.5%	PILOT [45]	High — prospective, international
VATS allocated vs received (unselected)	10 of 20	MIST-3 [37]	High — randomized, bias-corrected
Excluded on anaesthetic risk	2 of 20	MIST-3 [37]	High
LOS: IET / VATS / standard care	7 d / 7 d / 10 d, P=0.70 (3-group)	MIST-3 [37]	No significant difference between ANY arms; feasibility trial, not powered for LOS
EQ-5D at 2 mo, IET vs VATS	0.83 vs 0.59 (P=0.023)	MIST-3 [37]	Moderate — small n, secondary endpoint
12-mo mortality, VATS vs IET vs SOC	20% / 5.3% / 4.8% (P=0.19)	MIST-3 [37]	NS; ITT with 50% VATS compliance — not attributable to surgery
Lung cancer resections/yr US	~56,000	Potter [20]	High — counted
Any pleural adhesions	34% (49/144) – 41.4% (394/952)	Kouritas [21], Shiono [22]	[21] peer-reviewed; [22] abstract
Severe adhesions requiring a tool (>1 lobe)	11.8% (112/952)	Shiono [22]	Low — sole source, conference abstract
Rind formation pathway (MesoMT)	Characterized	Tucker & Idell [46]	High — supersedes 2004 [41]
Myofibroblast origin / persistence	Unresolved	Tucker & Idell [46]	High — basis for filter score of 3
tPA/DNase 5-dose course	~\$15,000	Kheir [32]	Moderate — single institution
Medical thoracoscopy	~\$9,700	Kheir [32]	Moderate
VATS	~\$20,000	Lacin [33]	Moderate
VATS-to-open 90-day cost gap	\$3,476	Farjah [15]	High figure; lobectomy proxy, driven by prolonged stay
Prolonged air leak cost	~\$2,556	Jones [16]	Pooled mean; lobectomy, not decortication
OR cost per minute	~\$26	Grossi [14]	Valve denominator, thoracic proxy
Fibrin sealant price point	~\$480/application	Zaraca [17]	Moderate
Pharma weighted-average COGS	25.7% (2021)	[34]	Moderate — industry aggregate
Medical vs surgical DRG	\$7–11K vs \$17–31K	CMS FY2025 IPPS [45]	High — weight tables verified directly
Inpatient cost per day	~\$2,800 derived; \$2,000–2,500 used	Derived from [1]	Moderate — average overstates marginal day
LOS reduction, device route	1 fewer day	Assumption	None — no clinical basis
Decortication volume, one surgeon	~1/week (~50/yr)	[Interview 1]	Single surgeon, single site; conflicts with STS per-institution rate [3]
Operative duration, observed case	90 min, VATS, no conversion	[Interview 1]	n=1 direct observation
Operative duration, matured rind	2–3 hours	[Interview 1]	Surgeon estimate, not measured
Composite procedure structure	Drainage + decortication + parietal pleurectomy	[Interview 1]	n=1; relevant to coding heterogeneity [43]

Appendix B — Glossary

Term	Definition
Decortication	Surgical removal of the fibrous peel restricting lung expansion (CPT 32220 total, 32225 partial) [43]
Empyema	Infected pleural effusion; pus in the pleural space
Fibrothorax	Diffuse pleural fibrosis producing a rigid, thickened pleura
IET / IPFT	Intrapleural enzymatic / fibrinolytic therapy; standard regimen is tPA plus DNase

Term	Definition
Lung entrapment	Failure of lung re-expansion due to an active, potentially reversible pleural process
Trapped lung	Chronic, stable non-expandable lung from mature fibrosis; a distinct entity from entrapment
Organizing rind / peel	Stage III collagenous membrane encasing the lung
PAI-1	Plasminogen activator inhibitor-1; principal inhibitor of pleural plasminogen activator activity
RAPID score	Renal, Age, Purulence, Infection source, Dietary — pleural infection risk score [10]
scuPA	Single-chain urokinase plasminogen activator; the LTI-01 molecule
NTAP	New Technology Add-on Payment; 3-year Medicare bridge outside the DRG
MesoMT	Mesomesenchymal transition; differentiation of pleural mesothelial cells into collagen-depositing myofibroblasts [46]
HILS	High Importance, Low Satisfaction; outcomes an executor values highly but existing solutions serve poorly
JTBD / ODI	Job to be done / outcome-driven innovation; method decomposing a job into steps and desired outcomes (Section 6)
Job map	The eight-step decomposition of a job to be done: Define, Locate, Prepare, Confirm, Execute, Monitor, Modify, Conclude
PAL	Prolonged air leak; air leak persisting beyond roughly five postoperative days. Occurs in 7.5% of empyema decortications [5]
FTO	Freedom to operate; analysis of whether a proposed design infringes existing patents
De novo	FDA pathway for a novel device with no suitable predicate, establishing a new classification
505(b)(2)	Abbreviated NDA pathway relying partly on prior approval data. Unavailable here: no pleural collagenase predicate exists
510(k)	FDA clearance pathway based on substantial equivalence to a legally marketed predicate device
BLA	Biologics License Application; the approval pathway required for the biologic route
J-code	HPCS code permitting separate payment for a drug; required for the biologic route
DRG	Diagnosis-Related Group; the fixed Medicare inpatient payment into which both routes are bundled
ASP	Average selling price
SAM	Serviceable addressable market; total value created multiplied by the share of it a purchaser will concede
Decortication vs pleurectomy	Decortication removes the peel from the visceral pleura (lung); parietal pleurectomy strips the lining from the chest wall. Separately coded [43]

Appendix C — Reference List

Primary interviews are cited in text as [Interview 1] (thoracic surgeon) and [Interview 2] (CTICU physician assistant). Numbered references follow.

- [1] Mumjadi SR, Edakkanambeth Varayil J, Bodilson A, et al. Epidemiology of Adult Pleural Disease in the United States. *Chest*. 2021;160(5):1534–1551. <https://pubmed.ncbi.nlm.nih.gov/34023322/>
- [2] Shen KR, Bribiesco A, Crabtree T, et al. The American Association for Thoracic Surgery consensus guidelines for the management of empyema. *J Thorac Cardiovasc Surg*. 2017;153(6):e129–e146. [https://www.jtcvs.org/article/S0022-5223\(17\)30152-6/pdf](https://www.jtcvs.org/article/S0022-5223(17)30152-6/pdf)
- [3] Towe CW, Carr SR, Donahue JM, et al. Morbidity and 30-day mortality after decortication for parapneumonic empyema and pleural effusion among patients in the Society of Thoracic Surgeons' General Thoracic Surgery Database. *J Thorac Cardiovasc Surg*. 2019;157(3):1288–1297. [https://www.jtcvs.org/article/S0022-5223\(18\)33151-9/fulltext](https://www.jtcvs.org/article/S0022-5223(18)33151-9/fulltext)
- [4] Bedawi EO, Ricciardi S, Hassan M, et al. ERS/ESTS statement on the management of pleural infection in adults. *Eur Respir J*. 2022;61(2):2201062. <https://publications.ersnet.org/content/erj/61/2/2201062>
- [5] Scheese D, et al. Prolonged air leak following decortication for empyema. *J Surg Res*. 2025;307:116–121. <https://pubmed.ncbi.nlm.nih.gov/40014907/>

- [6] Pulmonary function outcomes following decortication. Thoracic Research and Practice (formerly Turkish Thoracic Journal). 2019.
<https://thoracrespract.org/pdf/68ab204c-182b-49da-b227-bc7efe058632/articles/TurkThoracJ.2019.180146/6.pdf>
- [7] Endoh M, et al. Systematic review of empyema management (134 articles, 227,898 patients). Curr Challenges Thorac Surg. 2021. <https://ccts.amegroups.org/article/view/39521/pdf>
- [8] Arshava E. Empyema and Parapneumonic Effusions: General Considerations. Iowa Thoracic Surgery Protocols, University of Iowa. 2024. Institutional protocol page; not peer-reviewed.
<https://iowaprotocols.medicine.uiowa.edu/protocols/empyema-and-parapneumonic-effusions-general-considerations>
- [9] Beckert L, Brockway B, Simpson G, et al. Phase 1 trial of intrapleural LTI-01; single chain urokinase in complicated parapneumonic effusions or empyema. JCI Insight. 2019;5(10):e127470.
<https://pmc.ncbi.nlm.nih.gov/articles/PMC6542611/>
- [10] Lamb C, Akulian J, Musani A, et al. (Idell S, senior author). The LTI-01-2001 Phase 2 Trial of Intrapleural LTI-01 in Patients with Infected, Non-Draining Pleural Effusions. Research Square preprint, posted 10 April 2026; under review at Respiratory Research, revision requested 14 May 2026. ClinicalTrials.gov NCT04159831. doi:10.21203/rs.3.rs-9348224/v1. <https://www.researchsquare.com/article/rs-9348224/v1> — NOT PEER REVIEWED AT TIME OF WRITING
- [11] Tucker TA, Idell S, et al. Pleural fluid remodeling and organizing empyema biology. Clin Transl Med. 2016;5:29. <https://onlinelibrary.wiley.com/doi/full/10.1186/s40169-016-0097-2>
- [12] Reichert M, et al. Stage III empyema: surgical outcomes in 217 patients, 4.5% VATS conversion. 2018.
<https://pubmed.ncbi.nlm.nih.gov/29218675/>
- [13] Bender B, et al. VATS versus open thoracotomy for empyema: 53 patients, 44.4% conversion among VATS attempts. 2015. <https://pubmed.ncbi.nlm.nih.gov/25580760/>
- [14] Grossi EA, et al. Operating room cost per minute for thoracic and valve procedures, Premier multi-institution database. 2021. CAVEAT: valve-surgery denominator; thoracic-adjacent proxy, not decortication-measured. <https://pmc.ncbi.nlm.nih.gov/articles/PMC8341883/>
- [15] Farjah F, Backhus LM, Varghese TK, Mulligan MS, Cheng AM, Alfonso-Cristancho R, Flum DR, Wood DE. Ninety-day costs of video-assisted thoracic surgery versus open lobectomy for lung cancer. Ann Thorac Surg. 2014;98(1):191-196. PMID 24820393. MarketScan, 9,962 patients, 2007-2011; risk-adjusted 90-day costs \$3,476 lower for VATS (p=0.001), difference largely explained by prolonged length of stay. CAVEAT: lobectomy population; average approach gap, not measured cost of a conversion event.
[https://www.annalsthoracicsurgery.org/article/S0003-4975\(14\)00602-X/fulltext](https://www.annalsthoracicsurgery.org/article/S0003-4975(14)00602-X/fulltext)
- [16] Jones et al. Costs of adverse events in thoracic surgery: systematic review. 2023. PMID 37770272. Prolonged air leak ~\$2,556 (range \$571-3,573); pooled mean, more defensible than any single-center figure. CAVEAT: lobectomy data, not decortication. <https://pubmed.ncbi.nlm.nih.gov/37770272/>
- [17] Zaraca F, et al. Cost analysis of fibrin sealant use in thoracic surgery (Progel ~\$480/application). 2017.
<https://pubmed.ncbi.nlm.nih.gov/29312730/>
- [18] Rogers EM. Diffusion of Innovations. New York: Free Press; 1962.
- [19] Li SJ, et al. Presence of pleural adhesions can predict conversion to thoracotomy and postoperative surgical complications in patients undergoing VATS lung cancer lobectomy. J Thorac Dis. 2018;10(1):416-431.
<https://pubmed.ncbi.nlm.nih.gov/29600074/>
- [20] Potter AL, et al. US lung cancer resection volumes. Shanghai Chest. 2018.
<https://shc.amegroups.org/article/view/8191/html>

- [21] Kouritas VK, et al. Pleural adhesions in major lung resection: 49 of 144 patients (34%). *Interact Cardiovasc Thorac Surg*. 2017;25(4):613. <https://academic.oup.com/icvts/article/25/4/613/4031269>
- [22] Shiono S, Endo M, Nakahashi K. Clinical and Prognostic Impact of Pleural Adhesions during Lung Cancer Surgery. Abstract EP02.03-024, *J Thorac Oncol*. 2022;17(9 Suppl). doi:10.1016/j.jtho.2022.07.380. Yamagata Prefectural Central Hospital. Of 952 evaluated lobectomies: grade 0 (no adhesions) n=558; grade 1 (slight) n=282; grade 2 (involving >1 lobe) n=112 — 41.4% any adhesions, 11.8% severe. NOTE: pitch slide 11 credits this to Nakahashi; Shiono is first author. CAVEAT: conference e-poster abstract, not a peer-reviewed full paper; weaker evidence than the other prevalence sources. [https://www.jto.org/article/S1556-0864\(22\)00721-3/fulltext](https://www.jto.org/article/S1556-0864(22)00721-3/fulltext)
- [23] Operative versus nonoperative management of empyema, 9,014 hospitalized patients. *Ann Thorac Surg*. 2019. [https://www.annalsthoracicsurgery.org/article/S0003-4975\(19\)31059-8/fulltext](https://www.annalsthoracicsurgery.org/article/S0003-4975(19)31059-8/fulltext)
- [24] Operative versus nonoperative therapy for empyema: 30-day mortality, readmission, and discharge disposition. *J Thorac Cardiovasc Surg*. <https://www.sciencedirect.com/science/article/pii/S0022522306018617>
- [25] Primary empyema stage distribution: 81 patients, 36% Stage II, 64% Stage III. 2010. <https://pubmed.ncbi.nlm.nih.gov/20053696/>
- [26] Surgical treatment of Stage II–III empyema in 719 consecutive patients; conversion 8.4% Stage II vs 19.2% Stage III. <https://pmc.ncbi.nlm.nih.gov/articles/PMC9821231/>
- [27] Empyema stage distribution in 274 patients (Stage I/II/III = 78/157/39). <https://pmc.ncbi.nlm.nih.gov/articles/PMC4989967/>
- [28] Post-thoracotomy and post-VATS chronic pain; VATS conversion rates; unstratified tPA/DNase cure rate. Review. <https://pmc.ncbi.nlm.nih.gov/articles/PMC4466425/>
- [29] Rahman NM, Maskell NA, West A, et al. Intrapleural use of tissue plasminogen activator and DNase in pleural infection (MIST2). *N Engl J Med*. 2011;365(6):518–526. <https://www.nejm.org/doi/10.1056/NEJMoa1012740>
- [30] Khemasuwan D, Wilshire C, Reddy C, Gilbert C, Gorden J, Balwan A, Sanchez TM, Bixby B, Sorensen JS, Shojaaee S. Machine Learning Model Predictors of Intrapleural Tissue Plasminogen Activator and DNase Failure in Pleural Infection: A Multicenter Study. *Ann Am Thorac Soc*. 2025;22(2):187–192. doi:10.1513/AnnalsATS.202402-151OC. PMID 39471322. 466 patients, five US institutions; IET resolution 78% (n=365). Variable importance for pleural thickening: 36% in XGBoost and 18% in L1-penalized regression (ranked first in both), 26% in LightGBM and 13% in SVM (ranked second in both). Abscess/necrotizing pneumonia: 35% LightGBM, 17% SVM, 17% XGBoost. <https://pubmed.ncbi.nlm.nih.gov/39471322/>
- [31] Current limitations in predicting response to intrapleural fibrinolytic therapy versus surgery. *Front Pharmacol*. 2021;12:806393. <https://www.frontiersin.org/journals/pharmacology/articles/10.3389/fphar.2021.806393/full>
- [32] Kheir F, et al. Comparative cost of medical thoracoscopy (\$9,700) versus five doses of tPA/DNase (\$15,000). *Chest*. 2018. [https://journal.chestnet.org/article/S0012-3692\(18\)31666-0/fulltext](https://journal.chestnet.org/article/S0012-3692(18)31666-0/fulltext)
- [33] Lacin T, et al. Cost of VATS in thoracic surgery (~\$20,000). *J Thorac Dis*. 2013. <https://jtd.amegroups.org/article/view/1338/html>
- [34] Hardman & Co. Pharma statistics: long-term cost and underlying EBIT analysis. 2021. Weighted-average industry COGS 25.7%. <https://hardmanandco.com/2021-pharma-statistics-long-term-cost-underlying-ebit-analysis/>
- [35] Karpathiou G, et al. Histopathology of the organizing pleural rind. 2022. [Cited on pitch slide 4]

- [36] Akulian J, Bedawi EO, Abbas H, et al. Bleeding Risk With Combination Intrapleural Fibrinolytic and Enzyme Therapy in Pleural Infection: An International, Multicenter, Retrospective Cohort Study. *Chest*. 2022;162(6):1384–1392. doi:10.1016/j.chest.2022.06.008. PMID 35716828. 1,850 patients; bleeding complication rate 4.2%, no major adverse events. <https://pmc.ncbi.nlm.nih.gov/articles/PMC9773231/>
- [37] Bedawi EO, Stavroulias D, Hedley E, et al. Early Video-assisted Thoracoscopic Surgery or Intrapleural Enzyme Therapy in Pleural Infection: A Feasibility Randomized Controlled Trial. The Third Multicenter Intrapleural Sepsis Trial — MIST-3. *Am J Respir Crit Care Med*. 2023;208(12):1305–1315. doi:10.1164/rccm.202305-0854OC. PMID 37820359. Eight UK centers; 178 screened, 110 eligible, 60 randomized 1:1:1 to standard care (n=21), early IET (n=19), early VATS (n=20). Median time to intervention 1.0 vs 3.5 d (P=0.02). No significant difference in median LOS across the three arms: 7 d IET, 7 d VATS, 10 d standard care, 9 d overall, P=0.70 — the three-group comparison, so the apparent advantage over standard care was also non-significant. Not powered for LOS. Only 10/20 allocated to VATS received surgery; 2/20 excluded on anaesthetic risk. EQ-5D at 2 mo 0.83 (IET) vs 0.59 (VATS), P=0.023. ITT 12-month mortality: VATS 4/20 (20%), standard care 1/21 (4.8%), IET 1/19 (5.3%); chi-square 3.33 (df=2), P=0.19; 2 of 6 deaths occurred before discharge, both in the VATS arm. Open access. <https://pmc.ncbi.nlm.nih.gov/articles/PMC10765402/>
- [38] Siddiqui HA, Maginot ER, Moody TB, et al. Pleural Space Diseases and Their Management: What is the Role of Intrapleural Fibrinolytic Therapy? *Am Surg*. 2025;91:1036–1045. Reports intrapleural enzymatic treatment failure at up to 25%. NOTE: figure relayed via the LTI-01 Phase 2 report [10]
- [39] Idell S, Rahman NM. Intrapleural Fibrinolytic Therapy for Empyema and Pleural Loculation: Knowns and Unknowns. *Ann Am Thorac Soc*. 2018;15(5):515–517. doi:10.1513/AnnalsATS.201711-848PS. PMID 29361235. (Note: some indexes list this as 15(4); PubMed records 15(5).) Position statement: after ~70 years of IPFT, choice of agent, the role of fibrinolysin-plus-DNase combinations, and optimal dose and dosing interval all remain controversial and require rigorous evaluation; the authors call for mechanistic work enabling personalized therapy and bedside dosing guidance. <https://www.atsjournals.org/doi/10.1513/AnnalsATS.201711-848PS>
- [40] Komissarov AA, Florova G, Azghani AO, et al. Dose dependency of outcomes of intrapleural fibrinolytic therapy in new rabbit empyema models. *Am J Physiol Lung Cell Mol Physiol*. 2016;311(2):L389–399.
- [41] Mutsaers SE, Prele CM, Brody AR, Idell S. Pathogenesis of pleural fibrosis. *Respirology*. 2004;9(4):428–440. doi:10.1111/j.1440-1843.2004.00633.x. Defines pleural fibrosis as excessive matrix deposition destroying normal pleural architecture; identifies organized haemorrhagic effusion, tuberculous effusion, empyema, and asbestos-related pleurisy as converging etiologies; states explicitly that the pathogenesis is unknown and that the primary effector cell is contested between the subpleural fibroblast and the mesothelial cell. <https://onlinelibrary.wiley.com/doi/abs/10.1111/j.1440-1843.2004.00633.x>
- [42] Lee YC, Lane KB, Zoia O, et al. Transforming growth factor-beta induces collagen synthesis without inducing IL-8 production in mesothelial cells. *Eur Respir J*. 2003;22(2):197–202.
- [43] Tucker TA, Idell S. Update on Novel Targeted Therapy for Pleural Organization and Fibrosis. *Int J Mol Sci*. 2022;23(3):1587. doi:10.3390/ijms23031587. Characterizes mesomesenchymal transition (MesoMT): pleural mesothelial cells differentiate into myofibroblasts expressing alpha-SMA, collagen and fibronectin, losing E-cadherin and ZO-1; TGF-beta signaling via PI3K/Akt, NF-kappaB, GSK-3beta; PAI-1, DOCK2 and myocardin as participants. States that myofibroblast origin remains unclear and that how long and why they persist is not fully elucidated. Proposes blocking or reversing myofibroblast differentiation as a therapeutic target. <https://pmc.ncbi.nlm.nih.gov/articles/PMC8835949/>
- [44] Tucker TA, Komissarov AA, Idell S. Perspective and update: intrapleural fibrinolytic therapy for pleural infections and other forms of pleural organization. *Respir Res*. 2025. doi:10.1186/s12931-025-03184-y. <https://respiratory-research.biomedcentral.com/articles/10.1186/s12931-025-03184-y>

[45] CPT procedure descriptors for pulmonary decortication and pleurectomy: 32220 (total, open), 32225 (partial, open), 32310 (parietal pleurectomy), 32320 (decortication with parietal pleurectomy), 32651 (VATS, partial decortication), 32652 (VATS, total decortication with intrapleural pneumonolysis). AAPC code references, accessed July 2026.

[46] Centers for Medicare & Medicaid Services. FY2025 IPPS Final Rule, MS-DRG relative weight tables. Medical DRG for pleural infection reimburses approximately \$7,000–11,000 against \$17,000–31,000 for the surgical DRG. Weight tables downloaded and verified directly from the Final Rule by the author.

[47] Corcoran JP, Psallidas I, Gerry S, et al. Prospective validation of the RAPID clinical risk prediction score in adult patients with pleural infection: the PILOT study. *Eur Respir J.* 2020;56:2000130. doi:10.1183/13993003.00130-2020. PMID 32675200. Largest international prospective observational cohort in pleural infection; standard medical therapy with chest tube and antibiotics fails in 33.5% of cases.

[48] Complications of untreated empyema: pulmonary fibrosis with lung trapping, chest wall contraction and deformity, empyema necessitans, bronchopleural fistula, osteomyelitis, pericarditis, mediastinal and subphrenic abscess, sepsis, death. ScienceDirect topic review.
<https://www.sciencedirect.com/topics/medicine-and-dentistry/empyema>